Tianxiang XIA

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EDUCATION

ETH Zürich, Zürich, Switzerland — *since 09/2024*
Master's Degree, Major in Machine Intelligence — GPA Major: **6/6** Overall: 5.87 (in progress)

Coursework Full mark (6/6) in Probabilistic AI, Deep Learning, Algorithms Lab, Optimizations for Data Science, Big Data, Research in CS Memo, etc.

Sorbonne Université, Paris, France — *09/2021 - 06/2024*
Double Bachelor's Degree in Mathematics & Computer Science — 225 ECTS Grade : **19/20 (rank 1/70)**

Coursework - Mathematics: topology and multivariate calculus, functional analysis, linear and bilinear algebra, measure theory and probability, numerical analysis, algebra and arithmetic, etc.
- Computer Science: AI and operational research, numerical methods, OOP and functional programming, databases, algorithms, data structures, logic, OS, cryptography, networks, computer architecture, etc.

AWARDS AND HONORS

- ETH Zürich EESTEC AI Hackathon 2025, within a team of three students, **Second Prize** — *2024*
- ETH Zürich Viscon-19 Hackathon, within a team of three students, **First Prize** — *2024*
- International Collegiate Programming Contest, Southwestern Europe Regional Contest, **Bronze Medal** — *2024*
- Francilienne Programming Day (Journée Francilienne de Programmation) 12, **Rank 1/50** — *2022*
- 30^{th} China Team Selection Competition of International Olympiad in Informatics, **Silver Medal** — *2018*
- National Olympiad of Informatics in Provinces of China, **First Prize** — *2017*

SKILLS AND INTERESTS

Programming languages C/C++, Python, Java, OCaml, LaTeX, SQL, Intel/ARM Intrinsics

Spoken languages Chinese Mandarin (native), English (IELTS C1), French (C1, Bachelor's program in French)

Technical skills AI (PyTorch, vllm, LMCache), Unix-like System (Shell, Git, VIM), Docker, Web Development (HTML, CSS, JavaScript, React), Database Management (SQL, Spark, etc.)

Interests Large Models, Applied Mathematics, High-Performance & Large Scale Computing, Gym, Polyglot

WORK EXPERIENCE

Research Intern at ETH Zürich, SRI lab, ODI lab (present) — *since 03/2025*
- Redesigned reinforcement learning algorithms on multiple GPUs with verl for formal verification tasks
- Working on optimizations on manifold inspired Muon-variant LLM training methods
- Working on sparsity inducing training for LLM/VLM (RPCA, ADMM inspired)

Research Intern at Computing Product Lab, Huawei Zürich — *since 06/2025*
- Bringing high hit rate KV Cache system (report accepted as DAC 2026 WIP session poster) and online quantization to production ready in Multi LoRA Agent scenario.
- Contributing to open source communities PTO-Kernels, LMCache-Ascend, CANN/ops-nn

Teaching Assistant at Algorithms Lab (algolab), ETH Zürich — *09/2025 - 01/2026*
- In charge of oral assessment for students' performance in Master's level algorithms course

Research Intern at Huawei Paris with report & code — *06/2024 - 09/2024*
- Designed and built a neural symbolic reasoning engine on automatic Euclidean geometry theorem proving with human-computer interaction features through reasoning graph plotting and GeoGebra integration
- The project codebase was continued by HarmonicMath and won Gold Medal in IMO 2025

PROJECTS

Hardware Optimization for Clifford Nets with Report & code, at ETH Zürich — *03/2025 - 06/2025*
- Optimized the performance of Clifford neural layers with Intel and Apple M3 intrinsics ($\sim 20\%$ faster than PyTorch)

Automatic Neural Network Robustness Verification with code, ETH Zürich — *11/2024 - 12/2024*
- Implemented from scratch Deeppoly algorithms with learnable relaxation.
- Obtained the highest score in the final evaluation.

Text Representation Analysis with Slides, at LIP6, Sorbonne Université — *02/2023 - 06/2023*
- Self-studied and applied topological data analysis (representation by similarity filtration with time skeleton, analysis by persistent homology) to essay argument complexity assessment